Task 3 Report

Advanced SQL Analysis, KPI Modeling & Variance Dashboard

1. Introduction

The objective of this task is to analyze sales data using advanced SQL techniques, calculate key performance indicators, and develop a dashboard to understand monthly sales performance and business trends.

2. Tools Used

- SQL
- Microsoft Excel
- Microsoft Power BI

3. SQL Analysis

Advanced SQL queries were used to explore sales performance.

CASE Statement: Classified product categories into High, Medium, and Low Sales based on total sales.

HAVING Clause: Identified regions with total sales greater than the selected threshold.

Subquery: Compared category-wise sales with the average category sales.

Customer Analysis: Identified the top five customers based on total sales.

4. Excel Analysis

Monthly sales analysis was organized using the following metrics:

- Current Month Sales
- Previous Month Sales
- Sales Variance
- Growth Percentage

Sales variance was calculated by subtracting previous month sales from current month sales. Growth percentage was calculated by dividing sales variance by previous month sales and multiplying by 100.

5. Power BI Dashboard

The dashboard is designed to communicate monthly sales performance through:

- Current and previous month sales comparison
- Monthly sales variance
- Monthly growth percentage
- Key performance indicator cards

6. Business Insights

The SQL analysis identified Technology as the highest-selling category in the category-level results, with total sales of 827,455.87. Furniture and Office Supplies followed.

The top-customer analysis identified customers with the highest recorded sales, supporting customer performance analysis.

Monthly variance and growth analysis can help identify periods of increasing or decreasing sales and support business planning.

7. Data Limitation

The dataset does not contain a dedicated Profit column. Therefore, profit and profit-margin analysis could not be performed using the available fields.

8. Conclusion

This task provided practical experience in advanced SQL analysis, monthly KPI calculations, sales variance measurement, and business dashboard development. It demonstrates how data analysis tools can transform raw sales records into useful business information.